ShopSphere — E-commerce Shopping Funnel (DBMS Project Report)

Oct 14, 2026 · @pratheep

1. Project Title

ShopSphere: An E-commerce Shopping-Funnel Database with Behaviour Analytics

Database Management Systems (DBMS) course project, October 2026. Built with MySQL 8 and Django.

Course: 22AIE303 Database Management Systems.

GitHub repository: https://github.com/Bhavesh-max-jpj/E-commerce-project. It holds the source code, the database scripts (`database/`), documented queries, tests, screenshots and this report.

2. Team Members

#	Name	Roll number
1	Nagireddy Bhavesh Reddy	CB.SC.U4AIE24034
2	Bachu Harini	CB.SC.U4AIE24010
3	Kallam Pratheep Reddy	CB.SC.U4AIE24021
4	Ravala Gnanasri	CB.SC.U4AIE24064

3. Introduction and Problem Statement

ShopSphere is an online clothing and footwear store whose database records every step a shopper takes, from viewing a product to paying and reviewing it, so the store can see where customers drop off. It runs on Django with a MySQL 8 database (`shopping_funnel`) holding 2,905 products from 16 sellers.

Problem. A shop sees how many orders it gets, but not how many people looked and left. Without knowing where shoppers are lost (after viewing, in the cart, or at payment), it cannot fix the right step.

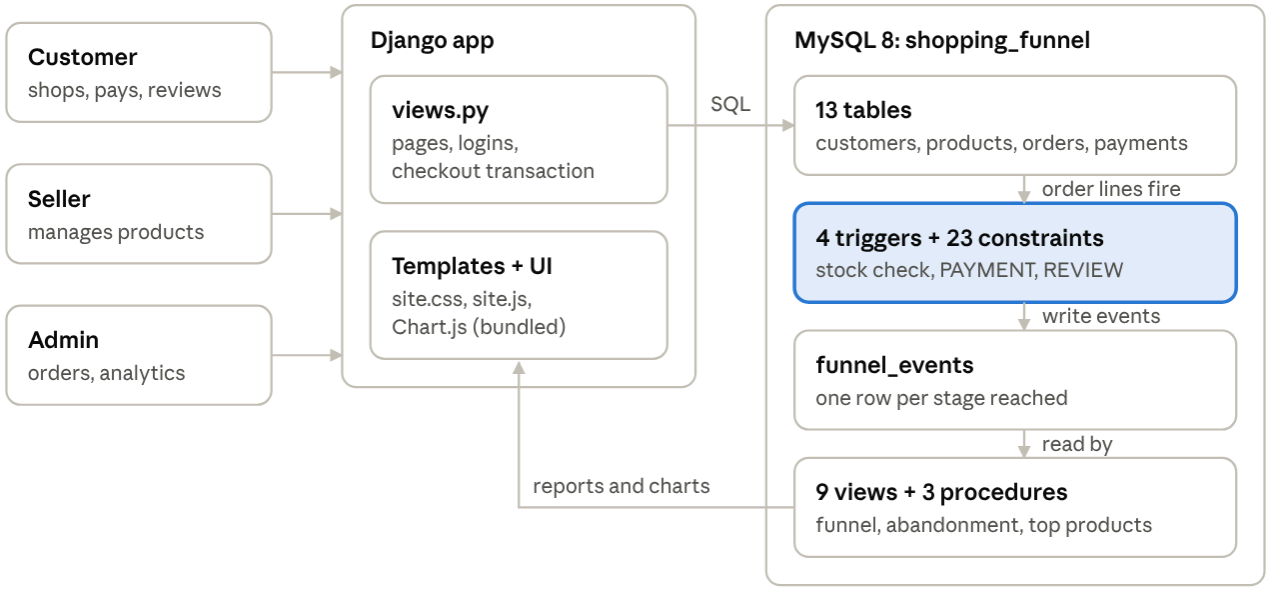
Approach. Each customer action is written to a `funnel_events` table. MySQL triggers, views and stored procedures then turn those events into a six-stage funnel (VIEW → CART → CHECKOUT → ORDER → PAYMENT → REVIEW) with conversion rates, drop-off, cart abandonment and breakdowns by product, category, brand, seller and device.

Main finding on the demo data. About 57% of visitors add something to the cart, but mobile shoppers complete only about 44% of their carts against about 60% on desktop, so mobile checkout is the weakest step.

System architecture

ShopSphere is a Django 5/6 web app (Python, PyMySQL) on a MySQL 8 database, with a hand-built HTML/CSS/JS interface. The key design choice: stock, funnel stages and data rules are enforced inside MySQL, so they hold whichever program writes to the database.

The database, not the web app, keeps stock and funnel data correct



system architecture · 3 roles, Django app, MySQL 8

Each role reaches its own pages; Django sends SQL; order lines fire the triggers, which write funnel events that the views and procedures turn into the admin's reports.

4. Conceptual and Logical Database Design

The conceptual model has 13 entities in three groups: the catalogue, the sales process and shopper behaviour. Each entity maps to exactly one table, and each one-to-many relationship becomes a foreign key on the "many" side.

Entities.

Group	Entity	Description	Identifier
Catalogue	Seller	A business that lists products	seller_id
Catalogue	Product	One item for sale, with price and stock	product_id
People	Customer	A registered shopper	customer_id
People	Admin	A store administrator	admin_id
Sales	Cart	A product waiting in a customer's cart	cart_id
Sales	Order	One purchase by a customer	order_id
Sales	Order item	One product line within an order (weak entity of Order)	order_item_id
Sales	Payment	A payment attempt for an order	payment_id
Sales	Review	A customer's rating of a product	review_id
Behaviour	User session	One login visit, with device type	session_id
Behaviour	Product view	A product page opened in a session	view_id
Behaviour	Cart event	A change to a cart (add, increase, decrease, remove)	event_id
Behaviour	Funnel event	A customer reaching a funnel stage	event_id

Relationships and cardinalities.

Relationship	Cardinality	Participation
Seller *lists* Product	1 : N	A product may have no seller (optional)
Customer *places* Order	1 : N	Every order has exactly one customer
Order *contains* Product, through Order item	M : N	An order has at least one item
Order *is paid by* Payment	1 : N	Failed attempts allowed; at most one completed
Customer *reviews* Product	M : N, through Review	Rating 1–5

Relationship	Cardinality	Participation
Customer *keeps* Product *in* Cart	M : N, through Cart	One row per customer and product
Customer *starts* User session	1 : N	
User session *includes* Product view	1 : N	
Cart *has history in* Cart event	1 : N	History kept when the cart row is deleted
Customer *reaches stage in* Funnel event	1 : N	Optional links to session, product and order

Logical design (ER to tables). Each strong entity becomes a table with a surrogate integer primary key. The two M : N relationships with their own attributes (Order item: quantity and price; Review: rating and text) become associative tables with two foreign keys. Optional relationships (product's seller; funnel event's session, product and order) use nullable foreign keys, so a funnel event survives if its product or session is later removed (`ON DELETE SET NULL`). The full table list is in section 9.

5. Normalization and Functional Dependencies

Every table is in BCNF except one stored derived column, `order_items.subtotal`, which is kept on purpose and guarded by a CHECK constraint. In every other table, the left side of each functional dependency (FD) is a candidate key.

Worked example: from one order sheet to 3NF. Start from a flat sheet of orders as a shop might keep it:

`ORDER_SHEET(order_id, order_date, customer_id, customer_name, city, {product_id, title, brand, price, quantity})`

1. **1NF.** The product list in braces is a repeating group. Moving it out gives one row per order line, with key (order_id, product_id).
2. **2NF.** With that composite key, `order_date, customer_id, customer_name, city` depend on `order_id` alone, and `title, brand, price` on `product_id` alone: partial dependencies. They are split into ORDERS, PRODUCTS and ORDER_ITEMS(order_id, product_id, quantity, unit_price).
3. **3NF.** In ORDERS, `customer_id → customer_name, city` is transitive (order_id → customer_id → name). Moving it out gives CUSTOMERS. The same step takes seller details out of PRODUCTS into SELLERS.

The result is exactly the design used: `customers`, `orders`, `order_items`, `products` and `sellers`.

Functional dependencies per table.

Table	Functional dependencies	Candidate keys	Normal form
sellers	seller_id → all; email → all	seller_id, email	BCNF
products	product_id → seller_id, gender, category, sub_category, product_type, colour, usage_type, product_title, image, image_url, brand, price, stock_quantity	product_id	BCNF
customers	customer_id → all; email → all	customer_id, email	BCNF
admin	admin_id → all; username → all; email → all	admin_id, username, email	BCNF
cart	cart_id → all; (customer_id, product_id) → cart_id, quantity, added_date	cart_id, (customer_id, product_id)	BCNF
orders	order_id → customer_id, order_date, total_amount, order_status, shipping_addr, delivery_date	order_id	BCNF
order_items	order_item_id → order_id, product_id, quantity, unit_price, subtotal; (quantity, unit_price) → subtotal	order_item_id	2NF (by design, see below)
payments	payment_id → all; transaction_id → all	payment_id, transaction_id	BCNF
reviews	review_id → customer_id, product_id, rating, review_text, review_date	review_id	BCNF
user_sessions	session_id → customer_id, start_time, end_time, device_type	session_id	BCNF

Table	Functional dependencies	Candidate keys	Normal form
product_views	view_id → customer_id, product_id, session_id, view_timestamp	view_id	BCNF
cart_events	event_id → customer_id, product_id, cart_id, event_type, event_timestamp	event_id	BCNF
funnel_events	event_id → customer_id, session_id, product_id, event_stage, event_timestamp, order_id	event_id	BCNF

Deliberate denormalization.

- `order_items.subtotal` depends on `quantity` and `unit_price`, which are not keys, so the table is strictly 2NF. It is stored so reports avoid recomputing it, and the CHECK `subtotal = quantity * unit_price` makes an inconsistent value impossible.
- `orders.total_amount` repeats the sum of the order's items. This is not an FD inside one table, but it is redundant; the clean-up script recalculated it, and the views use the item rows.
- `order_items.unit_price` is **not** redundant with `products.price`: it records the price at the moment of purchase, which later price changes must not alter.
- `funnel_events` repeats facts recorded elsewhere (views, carts, orders). It is an append-only log designed for fast analysis, the usual practice for event data.

6. Advanced SQL Implementation and Query Design

The database protects its own data. Bad values are refused by constraints, stock and funnel events are handled by triggers, and reports come from views and stored procedures. All of it lives in the numbered SQL scripts in the repository's `database/` folder (`01_schema.sql` to `09_delete_rules.sql`). The 24 documented complex queries are in `database/queries/complex_queries.sql`, and 16 SQL validation tests are in `database/tests/validation_tests.sql`.

Constraints (17 CHECK, 6 UNIQUE, plus all foreign keys).

Rule	Where
Quantity above 0; price, stock and totals 0 or more	cart, order_items, products, orders, payments

Rule	Where
`subtotal = quantity × unit_price`	order_items
Rating 1–5; seller rating 0–5	reviews, sellers
Only valid statuses (Pending, Paid, Shipped, Delivered, Cancelled; Completed, Failed, Refunded)	orders, payments
Only valid stages and cart event types	funnel_events, cart_events
Email must look like an email; gender from a fixed list	customers, sellers
One cart row per customer and product; unique transaction ids and emails	cart, payments, customers, sellers
Cart history is kept when a cart row is deleted (`ON DELETE SET NULL`)	cart_events → cart

Triggers.

Trigger	When	What it does
`trg_order_items_before_insert`	before an order line is saved	Locks the product row (`FOR UPDATE`) and raises `SIGNAL 45000` if stock is too low
`trg_order_items_after_insert`	after an order line is saved	Reduces `stock_quantity`
`trg_payments_after_insert`	after a successful payment	Writes a PAYMENT funnel event for each product in the order
`trg_reviews_after_insert`	after a review	Writes a REVIEW funnel event

Views. `vw_funnel_summary`, `vw_product_funnel`, `vw_category_funnel`, `vw_brand_funnel`, `vw_daily_funnel`, `vw_device_funnel`, `vw_cart_abandonment`, `vw_seller_sales` and the helper `vw_customer_product_reach`.

Stored procedures.

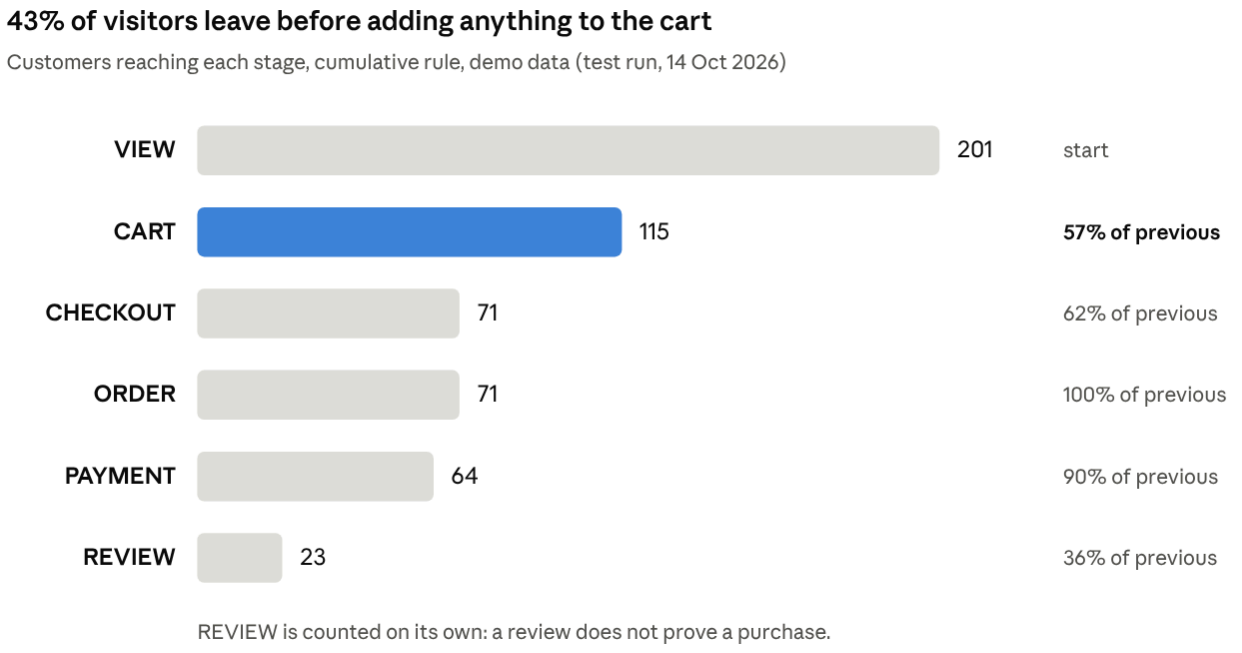
- `sp_funnel_between(start, end)`: the funnel for any date range. It is used by the analytics page's date filter.
- `sp_customer_journey(customer_id)`: one customer's steps in time order.
- `sp_top_products(n)`: best sellers by revenue from paid orders.

Transactions. Checkout runs inside one `transaction.atomic()` block: funnel events, the order, its lines and emptying the cart. If the stock trigger refuses any line, everything rolls back and the shopper sees "Not enough stock for …". Payment locks the order row so a double click cannot pay twice. Cancelling returns the stock and marks the payment Refunded in one transaction.

Funnel analytics query design

The funnel counts, for each stage, how many distinct customers got at least that far. On the demo data the biggest loss is the first step: 43% of visitors never add anything to the cart.

Cumulative rule. A customer who reached a later stage also counts in every earlier one (VIEW ⩾ CART ⩾ CHECKOUT ⩾ ORDER ⩾ PAYMENT), because nobody can pay without viewing and carting. In SQL, `MAX(FIELD(event_stage, 'VIEW', 'CART', 'CHECKOUT', 'ORDER', 'PAYMENT'))` finds each customer's furthest stage. Old orders placed before tracking existed therefore still give a sensible funnel. REVIEW is counted separately.



vw_funnel_summary on the test copy · 200 demo shoppers + test orders, 14 Oct 2026

Other findings on the same data.

- **Device:** mobile sessions turn 44% of carts into orders, against 60% on desktop and 67% on tablet. Mobile checkout is the weakest step.
- **Category:** footwear converts 54% of carts to payments, apparel 47%.
- **Abandonment:** 137 customer-and-product pairs were carted and never paid. `vw_cart_abandonment` lists them for follow-up.

Your own numbers will differ: they come from your MySQL and change as people shop. The admin Funnel Analytics page shows them live, with a date filter run by `sp_funnel_between`.

Key SQL code

The listings below are taken from the project's scripts (`database/04_triggers.sql`, `database/05_views_procedures.sql`, `database/08_cumulative_funnel.sql`). They show a trigger with locking and an error signal, a cumulative view, a stored procedure, and analysis queries using joins, grouping, subqueries and window functions.

Trigger: refuse an order line when stock is too low (`database/04_triggers.sql`)

```
CREATE TRIGGER trg_order_items_before_insert
BEFORE INSERT ON order_items
FOR EACH ROW
BEGIN
    DECLARE v_stock INT;
    DECLARE v_title VARCHAR(255);
    DECLARE v_msg   VARCHAR(255);

    IF NEW.quantity <= 0 THEN
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Quantity must be greater
than 0';
    END IF;

    SELECT stock_quantity, product_title INTO v_stock, v_title
      FROM products WHERE product_id = NEW.product_id
       FOR UPDATE;                      -- lock the row: two buyers cannot
take the last item

    IF v_stock IS NOT NULL AND v_stock < NEW.quantity THEN
        SET v_msg = CONCAT('Not enough stock for "', LEFT(v_title, 120), '":
only ', v_stock, ' left');
        SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = v_msg;
    END IF;
END
```

Trigger: record the PAYMENT stage automatically (`database/04_triggers.sql`)

```
CREATE TRIGGER trg_payments_after_insert
AFTER INSERT ON payments
FOR EACH ROW
BEGIN
    IF NEW.payment_status IN ('Completed', 'Successful', 'Paid') THEN
        INSERT INTO funnel_events (customer_id, session_id, product_id,
event_stage, event_timestamp, order_id)
        SELECT o.customer_id,
               (SELECT us.session_id FROM user_sessions us
                 WHERE us.customer_id = o.customer_id ORDER BY us.session_id
DESC LIMIT 1),
               oi.product_id, 'PAYMENT', COALESCE(NEW.payment_date, NOW()),
o.order_id
        FROM orders o JOIN order_items oi ON oi.order_id = o.order_id
        WHERE o.order_id = NEW.order_id;
    END IF;
END
```

View: cumulative funnel per product (`database/08_cumulative_funnel.sql`)

```
CREATE OR REPLACE VIEW vw_customer_product_reach AS
SELECT customer_id, product_id,
       MAX(FIELD(event_stage, 'VIEW','CART','CHECKOUT','ORDER','PAYMENT')) AS
max_rank,
       MAX(event_stage = 'REVIEW') AS reviewed
FROM funnel_events
WHERE product_id IS NOT NULL
GROUP BY customer_id, product_id;

CREATE OR REPLACE VIEW vw_product_funnel AS
SELECT p.product_id, p.product_title, p.brand,
       SUM(r.max_rank >= 1) AS viewed,   SUM(r.max_rank >= 2) AS carted,
       SUM(r.max_rank >= 3) AS checked_out, SUM(r.max_rank >= 4) AS ordered,
       SUM(r.max_rank >= 5) AS paid,     SUM(r.reviewed)      AS reviewed,
       ROUND(100 * SUM(r.max_rank >= 5) / NULLIF(SUM(r.max_rank >= 1), 0), 1)
AS view_to_paid_pct
FROM vw_customer_product_reach r
JOIN products p ON p.product_id = r.product_id
GROUP BY p.product_id, p.product_title, p.brand;
```

Stored procedure: funnel for a date range, called by the admin page's date filter

```
CALL sp_funnel_between('2026-09-15', '2026-10-14');
```

Inside, a CTE finds each customer's furthest stage in the range, and `LAG(customers) OVER (ORDER BY stage_order)` computes the percentage kept from the previous stage.

Analysis queries.

```
-- 1. Where are shoppers lost? Stages ordered by the share kept from the
previous stage
SELECT stage, customers, pct_from_previous
FROM vw_funnel_summary
WHERE pct_from_previous IS NOT NULL
ORDER BY pct_from_previous;

-- 2. Mobile against desktop: share of carts that became orders
SELECT device_type, sessions, carted, ordered, cart_to_order_pct
FROM vw_device_funnel
ORDER BY cart_to_order_pct;

-- 3. Money left in abandoned carts, by category (join + group + having)
SELECT p.category, COUNT(*) AS abandoned_items, SUM(a.price) AS value_lost
FROM vw_cart_abandonment a
JOIN products p ON p.product_id = a.product_id
GROUP BY p.category
HAVING COUNT(*) >= 5
ORDER BY value_lost DESC;

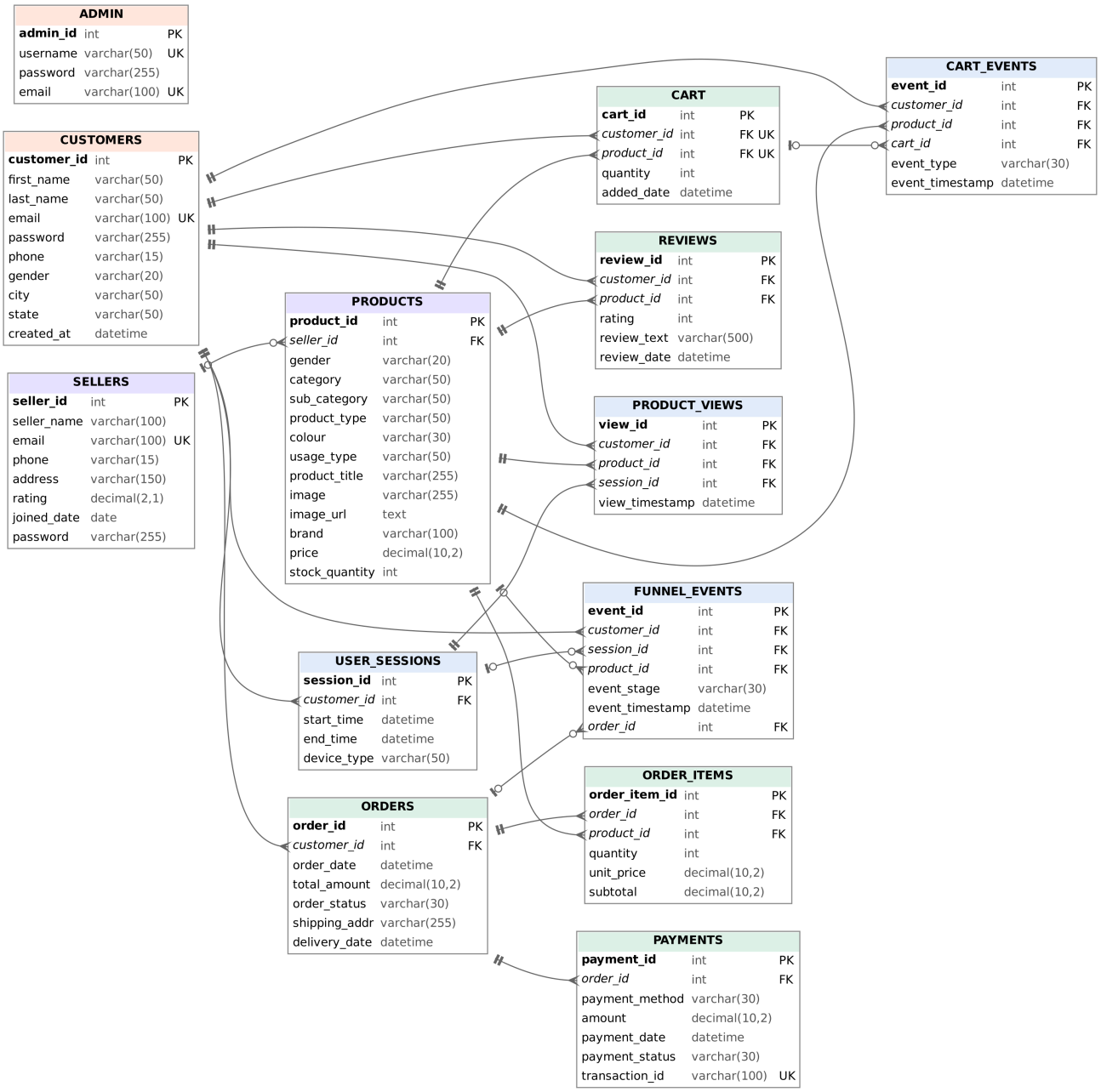
-- 4. Sellers earning more than the average seller (subquery)
SELECT seller_name, revenue
FROM vw_seller_sales
WHERE revenue > (SELECT AVG(revenue) FROM vw_seller_sales)
ORDER BY revenue DESC;

-- 5. Each customer's spending rank (window function)
SELECT c.first_name, c.last_name, SUM(o.total_amount) AS spent,
       RANK() OVER (ORDER BY SUM(o.total_amount) DESC) AS spend_rank
FROM customers c
JOIN orders o ON o.customer_id = c.customer_id
WHERE o.order_status IN ('Paid', 'Shipped', 'Delivered')
GROUP BY c.customer_id, c.first_name, c.last_name
ORDER BY spend_rank
LIMIT 10;
```

7. ER Diagram and UML Diagram

ER diagram

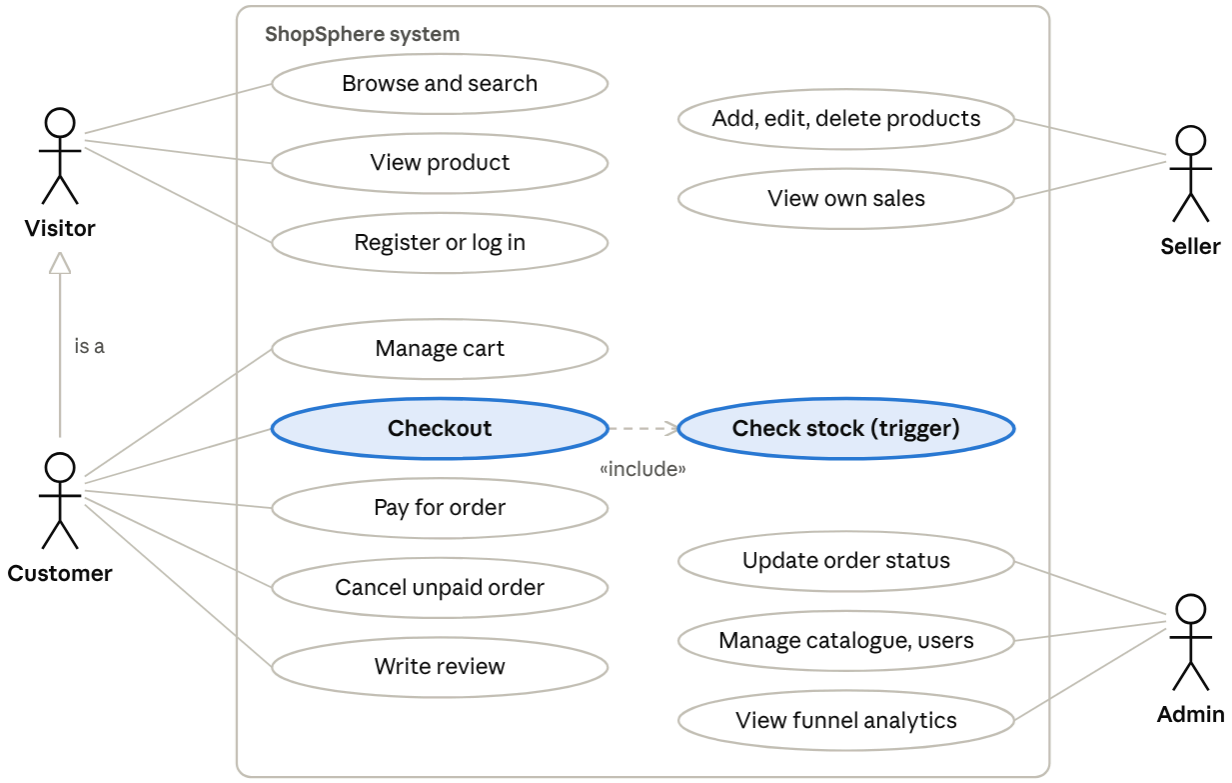
The ER diagram shows all 13 tables with their primary keys (PK), foreign keys (FK) and unique keys (UK). Crow's-foot notation: the three-pronged end marks the "many" side, a double bar marks "exactly one", and a circle marks "optional" (a nullable foreign key). Primary keys are bold, foreign keys italic. The full-resolution image is in the project at docs/er_diagram.png.



Customers and Products sit at the centre: almost every sales and behaviour table links to both. Funnel events link optionally to sessions, products and orders.

UML use-case diagram

Three user roles share one database; checkout always runs the stock trigger



UML use-case diagram · 4 actors, 14 use cases

A Customer is a Visitor who has logged in, so the customer inherits browsing and search. Checkout always includes the stock check, which runs inside MySQL as a trigger rather than in the web app.

8. Relational Schema

The schema has 13 relations. **Bold** marks the primary key and *italic* marks a foreign key; each foreign key's target is listed under its relation. Alternate keys (UNIQUE): SELLERS.email, CUSTOMERS.email, ADMIN.username, ADMIN.email, CART(customer_id, product_id), PAYMENTS.transaction_id.

- **SELLERS**(**seller_id**, seller_name, email, phone, address, rating, joined_date, password)
- **PRODUCTS**(**product_id**, *seller_id*, gender, category, sub_category, product_type, colour, usage_type, product_title, image, image_url, brand, price, stock_quantity) references: *seller_id* → SELLERS
- **CUSTOMERS**(**customer_id**, first_name, last_name, email, password, phone, gender, city, state, created_at)
- **ADMIN**(**admin_id**, username, password, email)

- **USER_SESSIONS**(**session_id**, *customer_id*, start_time, end_time, device_type)
  references: *customer_id* → CUSTOMERS
- **PRODUCT_VIEWS**(**view_id**, *customer_id*, *product_id*, *session_id*, view_timestamp)
  references: *customer_id* → CUSTOMERS; *product_id* → PRODUCTS; *session_id* → USER_SESSIONS
- **CART**(**cart_id**, *customer_id*, *product_id*, quantity, added_date)
  references: *customer_id* → CUSTOMERS; *product_id* → PRODUCTS
- **CART_EVENTS**(**event_id**, *customer_id*, *product_id*, *cart_id*, event_type, event_timestamp)
  references: *customer_id* → CUSTOMERS; *product_id* → PRODUCTS; *cart_id* → CART (ON DELETE SET NULL)
- **ORDERS**(**order_id**, *customer_id*, order_date, total_amount, order_status, shipping_addr, delivery_date)
  references: *customer_id* → CUSTOMERS
- **ORDER_ITEMS**(**order_item_id**, *order_id*, *product_id*, quantity, unit_price, subtotal)
  references: *order_id* → ORDERS; *product_id* → PRODUCTS
- **PAYMENTS**(**payment_id**, *order_id*, payment_method, amount, payment_date, payment_status, transaction_id)
  references: *order_id* → ORDERS
- **REVIEWS**(**review_id**, *customer_id*, *product_id*, rating, review_text, review_date)
  references: *customer_id* → CUSTOMERS; *product_id* → PRODUCTS
- **FUNNEL_EVENTS**(**event_id**, *customer_id*, *session_id*, *product_id*, event_stage, event_timestamp, *order_id*)
  references: *customer_id* → CUSTOMERS; *session_id* → USER_SESSIONS, *product_id* → PRODUCTS, *order_id* → ORDERS (all three ON DELETE SET NULL)

9. Database Tables

The 13 tables below are given with every attribute, its type, whether it may be NULL, its keys and defaults, its CHECK constraints and sample records. Structures are as in the final database. Samples come from the test copy, whose customers are generated demo shoppers; passwords are stored as Django `pbkdf2_sha256` hashes, shortened here.

9.1 sellers

Businesses that list products. Rows: 16.

Attribute	Type	Null	Key	Default / extra
seller_id	INT	no	PK	—
seller_name	VARCHAR(100)	no	—	—
email	VARCHAR(100)	no	UNIQUE	—
phone	VARCHAR(15)	yes	—	—
address	VARCHAR(150)	yes	—	—
rating	DECIMAL(2,1)	yes	—	—
joined_date	DATE	yes	—	—
password	VARCHAR(255)	yes	—	—

CHECK constraints: `chk_sellers_email`, `chk_sellers_rating`.

Sample records:

seller_id	seller_name	email	phone	address	rating	joined_date	password
1	TrendStyle Fashion	trendstyle@gmail.com	9876543210	Bengaluru, Karnataka	4.8	2025-01-10	pbkdf2_sha256$...
2	Urban Wear Hub	urbanwear@gmail.com	9876543211	Chennai, Tamil Nadu	4.6	2025-01-15	NULL
3	Elite Apparel	eliteapparel@gmail.com	9876543212	Hyderabad, Telangana	4.7	2025-01-20	NULL

9.2 products

The catalogue: one row per product. Rows: 2,905.

Attribute	Type	Null	Key	Default / extra
product_id	INT	no	PK	—
seller_id	INT	yes	FK → sellers(seller_id)	—
gender	VARCHAR(20)	yes	—	—
category	VARCHAR(50)	yes	—	—
sub_category	VARCHAR(50)	yes	—	—
product_type	VARCHAR(50)	yes	—	—
colour	VARCHAR(30)	yes	—	—

Attribute	Type	Null	Key	Default / extra
usage_type	VARCHAR(50)	yes	—	—
product_title	VARCHAR(255)	yes	—	—
image	VARCHAR(255)	yes	—	—
image_url	TEXT	yes	—	—
brand	VARCHAR(100)	yes	—	—
price	DECIMAL(10,2)	yes	—	—
stock_quantity	INT	yes	—	—

CHECK constraints: `chk_products_price`, `chk_products_stock`.

Sample records (key columns):

product_id	seller_id	gender	category	product_type	product_title	brand	price	stock_quantity
1636	10	Men	Footwear	Sports Shoes	Nike Men Air Zoom Century Shoes	Nike	3701.00	60
10096	12	Women	Footwear	Sports Shoes	Reebok Women Reeflex Run Blue Sports Shoes	Reebok	1834.00	142
12844	2	Boys	Apparel	Tshirts	Ant Kids Boys Multicoloured Striped T-shirts	Ant	4696.00	19

9.3 customers

Registered shoppers.

Attribute	Type	Null	Key	Default / extra
customer_id	INT	no	PK	AUTO_INCREMENT
first_name	VARCHAR(50)	no	—	—
last_name	VARCHAR(50)	no	—	—
email	VARCHAR(100)	no	UNIQUE	—
password	VARCHAR(255)	no	—	—
phone	VARCHAR(15)	yes	—	—
gender	VARCHAR(20)	yes	—	—
city	VARCHAR(50)	yes	—	—

Attribute	Type	Null	Key	Default / extra
state	VARCHAR(50)	yes	—	—
created_at	DATETIME	yes	—	CURRENT_TIMESTAMP

CHECK constraints: `chk_customers_email`, `chk_customers_gender`.

Sample records (key columns):

customer_id	first_name	last_name	email	gender	city	state	created_at
1	Sai	Pillai	sai.pillai1@demo.shop	Other	Kochi	Kerala	2026-09-25
2	Aarav	Iyer	aarav.iyer2@demo.shop	Female	Chennai	Tamil Nadu	2026-09-26
3	Divya	Das	divya.das3@demo.shop	Male	Coimbatore	Tamil Nadu	2026-09-30

9.4 admin

Store administrators who log in to the admin console.

Attribute	Type	Null	Key	Default / extra
admin_id	INT	no	PK	AUTO_INCREMENT
username	VARCHAR(50)	no	UNIQUE	—
password	VARCHAR(255)	no	—	—
email	VARCHAR(100)	no	UNIQUE	—

Sample record:

admin_id	username	password	email
1	admin	pbkdf2_sha256$…	admin@example.com

9.5 user_sessions

One row per customer login session.

Attribute	Type	Null	Key	Default / extra
session_id	INT	no	PK	AUTO_INCREMENT

Attribute	Type	Null	Key	Default / extra
customer_id	INT	no	FK → customers(customer_id)	—
start_time	DATETIME	yes	—	CURRENT_TIMESTAMP
end_time	DATETIME	yes	—	set at logout
device_type	VARCHAR(50)	yes	—	—

Sample records:

session_id	customer_id	start_time	end_time	device_type
1	1	2026-09-25 11:48	2026-09-25 12:04	Mobile
2	2	2026-09-26 11:37	2026-09-26 11:55	Tablet
3	2	2026-09-28 22:18	2026-09-28 22:44	Desktop

9.6 product_views

Every product page a logged-in customer opens.

Attribute	Type	Null	Key	Default / extra
view_id	INT	no	PK	AUTO_INCREMENT
customer_id	INT	no	FK → customers(customer_id)	—
product_id	INT	no	FK → products(product_id)	—
session_id	INT	no	FK → user_sessions(session_id)	—
view_timestamp	DATETIME	yes	—	CURRENT_TIMESTAMP

Sample records:

view_id	customer_id	product_id	session_id	view_timestamp
2427	201	6628	410	2026-10-14 12:34
2428	201	24260	410	2026-10-14 12:34
2429	201	3567	410	2026-10-14 12:34

9.7 cart

Items currently in a customer's cart.

Attribute	Type	Null	Key	Default / extra
cart_id	INT	no	PK	AUTO_INCREMENT
customer_id	INT	no	FK → customers(customer_id); UNIQUE with product_id	—
product_id	INT	no	FK → products(product_id); UNIQUE with customer_id	—
quantity	INT	no	—	—
added_date	DATETIME	yes	—	CURRENT_TIMESTAMP

CHECK constraints: `chk_cart_quantity`. Unique key: `uq_cart_customer_product`.

Sample records:

cart_id	customer_id	product_id	quantity	added_date
257	197	43883	1	2026-09-16 14:49
258	198	28606	1	2026-10-03 03:04
259	198	34858	1	2026-10-03 03:07

9.8 cart_events

History of every cart change; kept when the cart row is deleted.

Attribute	Type	Null	Key	Default / extra
event_id	INT	no	PK	AUTO_INCREMENT
customer_id	INT	no	FK → customers(customer_id)	—
product_id	INT	no	FK → products(product_id)	—
cart_id	INT	yes	FK → cart(cart_id), ON DELETE SET NULL	—
event_type	VARCHAR(30)	no	—	—
event_timestamp	DATETIME	yes	—	CURRENT_TIMESTAMP

CHECK constraints: `chk_cart_event_type` (ADD, INCREASE, DECREASE, REMOVE).

Sample records (cart_id is NULL because these cart rows were later checked out or removed):

event_id	customer_id	product_id	cart_id	event_type	event_timestamp
270	201	1637	NULL	INCREASE	2026-10-14 08:43
271	201	1637	NULL	DECREASE	2026-10-14 08:43
272	201	1637	NULL	REMOVE	2026-10-14 08:43
281	201	10096	NULL	ADD	2026-10-14 12:33

9.9 orders

One row per order placed.

Attribute	Type	Null	Key	Default / extra
order_id	INT	no	PK	AUTO_INCREMENT
customer_id	INT	no	FK → customers(customer_id)	—
order_date	DATETIME	yes	—	CURRENT_TIMESTAMP
total_amount	DECIMAL(10,2)	no	—	—
order_status	VARCHAR(30)	yes	—	Pending
shipping_addr	VARCHAR(255)	yes	—	—
delivery_date	DATETIME	yes	—	—

CHECK constraints: `chk_orders_status`, `chk_orders_total`.

Sample records:

order_id	customer_id	order_date	total_amount	order_status	shipping_addr	delivery_date
1	3	2026-09-30 20:39	6974.00	Paid	84, Main Road, Coimbatore, Tamil Nadu	2026-10-14
2	10	2026-10-07 22:44	6299.00	Paid	65, Main Road, Coimbatore, Tamil Nadu	2026-10-12
3	13	2026-09-19 05:21	7468.00	Paid	117, Main Road, Ahmedabad, Gujarat	2026-09-23

9.10 order_items

One row per product line in an order.

Attribute	Type	Null	Key	Default / extra
order_item_id	INT	no	PK	AUTO_INCREMENT
order_id	INT	no	FK → orders(order_id)	—
product_id	INT	no	FK → products(product_id)	—
quantity	INT	no	—	—
unit_price	DECIMAL(10,2)	no	—	—
subtotal	DECIMAL(10,2)	no	—	—

CHECK constraints: `chk_items_price`, `chk_items_quantity`, `chk_items_subtotal`.
Triggers: `trg_order_items_before_insert`, `trg_order_items_after_insert`.

Sample records:

order_item_id	order_id	product_id	quantity	unit_price	subtotal
1	1	31160	2	3487.00	6974.00
2	2	36957	2	2830.00	5660.00
3	2	50776	1	639.00	639.00

9.11 payments

Payments made against orders.

Attribute	Type	Null	Key	Default / extra
payment_id	INT	no	PK	AUTO_INCREMENT
order_id	INT	no	FK → orders(order_id)	—
payment_method	VARCHAR(30)	no	—	—
amount	DECIMAL(10,2)	no	—	—
payment_date	DATETIME	yes	—	CURRENT_TIMESTAMP
payment_status	VARCHAR(30)	yes	—	Pending

Attribute	Type	Null	Key	Default / extra
transaction_id	VARCHAR(100)	yes	UNIQUE	—

CHECK constraints: `chk_payments_amount`, `chk_payments_status`. Trigger: `trg_payments_after_insert`.

Sample records:

payment_id	order_id	payment_method	amount	payment_date	payment_status	transaction_id
1	1	COD	6974.00	2026-09-30 20:41	Completed	DEMO1506637
2	2	Card	6299.00	2026-10-07 22:45	Completed	DEMO2154302
3	3	NetBanking	7468.00	2026-09-19 05:21	Completed	DEMO3691194

9.12 reviews

Product ratings and comments.

Attribute	Type	Null	Key	Default / extra
review_id	INT	no	PK	AUTO_INCREMENT
customer_id	INT	no	FK → customers(customer_id)	—
product_id	INT	no	FK → products(product_id)	—
rating	INT	no	—	—
review_text	VARCHAR(500)	yes	—	—
review_date	DATETIME	yes	—	CURRENT_TIMESTAMP

CHECK constraints: `chk_review_rating` (1 to 5). Trigger: `trg_reviews_after_insert`.

Sample records:

review_id	customer_id	product_id	rating	review_text	review_date
1	13	26598	4	Nice, but size runs slightly small.	2026-09-28
2	14	40136	3	Average. Okay for daily use.	2026-10-01
3	19	5911	5	Great product, fast delivery.	2026-09-30

9.13 funnel_events

One row each time a customer reaches a funnel stage.

Attribute	Type	Null	Key	Default / extra
event_id	INT	no	PK	AUTO_INCREMENT
customer_id	INT	no	FK → customers(customer_id)	—
session_id	INT	yes	FK → user_sessions(session_id), ON DELETE SET NULL	—
product_id	INT	yes	FK → products(product_id), ON DELETE SET NULL	—
event_stage	VARCHAR(30)	no	index	—
event_timestamp	DATETIME	no	index	CURRENT_TIMESTAMP
order_id	INT	yes	FK → orders(order_id), ON DELETE SET NULL	—

CHECK constraints: `chk_funnel_stage` (VIEW, CART, CHECKOUT, ORDER, PAYMENT, REVIEW). Indexes: (event_stage, customer_id), (product_id, event_stage), (customer_id, event_timestamp), (event_timestamp).

Sample records: one demo customer's complete journey for one product, from viewing to reviewing:

event_id	customer_id	session_id	product_id	event_stage	event_timestamp	order_id
160	13	27	26598	VIEW	2026-09-19 05:07	NULL
161	13	27	26598	CART	2026-09-19 05:08	NULL
165	13	27	26598	CHECKOUT	2026-09-19 05:21	NULL
168	13	27	26598	ORDER	2026-09-19 05:21	3
171	13	27	26598	PAYMENT	2026-09-19 05:21	3
173	13	27	26598	REVIEW	2026-09-28 05:21	NULL

10. Conclusion and Future Scope

The project is complete for a DBMS course: every planned feature works and was tested. The items below are known limits, not unfinished work.

Limitations.

- Payment is simulated; no money moves.
- Views and cart adds are recorded only for logged-in customers, because the funnel needs to know who the shopper is.

- Orders placed before tracking began (August 2026) have no recorded views or cart adds. The cumulative funnel rule counts those customers in the earlier stages.
- The demo shoppers are generated, so their percentages show how the reports work rather than real market behaviour.

Future work. Guest carts that merge into the account on login, real payment gateway, email notifications on shipping, product photo upload by sellers, and indexes tuned with `EXPLAIN` on larger data.

Conclusion. ShopSphere shows how a database can do more than store data: constraints refuse bad values, triggers manage stock and record funnel stages, transactions keep orders whole, and views and stored procedures turn raw events into decisions. On the demo data, the clearest decision is to fix mobile checkout, where carts convert about 16 points worse than on desktop. Testing evidence, the operation checklist and setup steps are in the Appendix tab.

Appendix: features, testing and setup

Material that supports the report but sits outside the ten required sections.

A. Features by role

The site has three kinds of user, each with its own login page and pages. A login of the wrong kind (say, a customer on the seller page) is refused with a message.

Role	Can do
Visitor	Browse 2,905 products with search, filters (Men, Women, Boys, Girls; Footwear, Apparel), sorting and 24 per page; open a product; register
Customer	Add to cart, change quantity, remove; Buy now; checkout with an address; pay (UPI, Card, Net banking, Cash on delivery, simulated); see orders and their progress; cancel an unpaid order; write reviews with 1–5 stars; view profile
Seller	See products listed, units sold and revenue; search own products; add, edit and delete products (a product with order history is taken off sale instead of deleted)
Admin	Overview of revenue, orders, customers, products, recent orders and low stock; manage products; search customers; seller sales; manage orders (mark Shipped and Delivered, cancel); Funnel Analytics with date range, charts and reports; funnel per product

Interface (ShopSphere design).

- One stylesheet (`store/static/store/css/site.css`) and one script (`site.js`) shared by every page.
- 3D effects: a rotating ring of products on the home page, product cards that tilt toward the mouse with a light reflection, floating tilted login cards, tilting stat tiles, and a payment card that flips on hover.
- Light and dark mode with a switch in the header, remembered per browser.
- Phone layout with a two-column product grid and a collapsible menu.
- Animations turn off for visitors whose computer asks for reduced motion.

- Product photos come from the project's `datasets` folder (all 2,905 are there), so the site works offline. Chart.js is bundled for the same reason.

B. Security and bug fixes from the audit

The audit found two serious security holes and more than a dozen faults. All are fixed and re-tested.

Severity	Problem found	Fix
Critical	Registering a customer with the email `admin` (or a seller's email) reset that account's password, so anyone could take over the admin or a seller	Registration refuses any email already used by any login, checks the email format, and creates the login and the customer together in one transaction
Critical	A customer could open and pay someone else's order, and pay the same order twice	Payment pages only open the customer's own unpaid order, and the order row is locked while paying
High	`funnel_events` table was missing, so funnel tracking failed silently and the admin funnel page crashed	Table created, with history rebuilt from existing data
High	Plain-text passwords stored for 6 customers	Hashed with `python manage.py hash_plain_passwords`
High	Checkout saved orders in separate steps, so a failure left half-saved orders	One transaction; the stock trigger rolls the whole order back
Medium	11 crashes: unknown cart or product ids, rating text, missing address, duplicate product id, text in price, deleting a product with orders	Each now shows a clear message or redirects
Medium	The cart's Remove button always crashed (foreign key)	`cart_events` keeps history with `ON DELETE SET NULL`
Medium	Order totals, statuses and dates inconsistent (for example order 1 total ₹1,598 against items ₹9,436)	Corrected by `database/06_constraints_cleanup.sql`, and CHECK rules stop it recurring

Severity	Problem found	Fix
Low	Logging in from a product page lost your place	Every login and registration returns you to the page you were on; unsafe outside links are ignored
Low	No way to ship, deliver or cancel orders, and session end times never recorded	Admin status buttons, customer cancel with stock returned, and logout records the end time
Low	Database password stored in `settings.py` on GitHub	Moved to `config/local_settings.py`, which is excluded from Git

C. Final verification

On 14 October 2026 the whole project passed every automated check on a copy of the database rebuilt from the SQL scripts. No page crashes, no broken links, and no redirect loops.

Check	Result
Rebuild from empty: 9 SQL scripts + `migrate` + sample accounts + 200 demo shoppers	Passed
Django system check	0 issues
Every page and form as visitor, customer, seller and admin, including bad input	0 crashes
Link crawl: every link, image and form target on 864 pages across the four roles	0 broken, 0 loops
Full purchase: login, view, add, change quantity, remove, checkout, pay	Passed; stock fell by the quantity bought
Stock trigger: ordering more than stock	Refused; order, items and cart rolled back
Account takeover attempts (register as `admin` or a seller's email)	Blocked; real passwords still work
Paying another customer's order	Refused (404)
Order flow: Paid → Shipped → Delivered; illegal jumps; cancel	Only legal steps allowed; cancel returns stock and refunds payment

Check	Result
Login return: anonymous Add to cart → login → back to the item	Passed
Funnel views: every product funnel never rises from one stage to the next	Passed
Desktop, phone (390 px) and dark mode screenshots	No overflow, no sideways scroll

Automated tests in the repository.

Test suite	How to run	Result
29 Django application tests (`store/tests.py`): constraints, accounts and security, the whole funnel, stock rollback, seller and admin pages. A separate `test_shopping_funnel` database is built from `database/01` – `09` for them.	`python manage.py test store -v 2`	29 of 29 passed
16 SQL validation tests (`database/tests/validation_tests.sql`): CHECK, UNIQUE, NOT NULL, foreign keys, the 4 triggers and the ON DELETE rules, all rolled back afterwards	Execute All in MySQL Workbench	16 of 16 passed
24 documented complex queries (`database/queries/complex_queries.sql`)	Execute All in MySQL Workbench	All return results; outputs in `docs/sample_outputs/`

Step-by-step manual test cases for every page and role are in `docs/TEST_CASES.md`, and 14 screenshots of the running site are in `docs/screenshots/`.

D. Test checklist: every operation

Work through these in order with the site running at http://127.0.0.1:8000. Each line says what to do and what you should see; tick it when it passes.

Database (MySQL Workbench)

- ☐ `SHOW TRIGGERS FROM shopping_funnel;` → 4 triggers
- ☐ `SELECT * FROM vw_funnel_summary;` → 6 stages, each count no higher than the one above (REVIEW aside)

- [ ] `CALL sp_funnel_between('2026-09-01', '2026-10-31');` → the funnel for those dates
- [ ] `CALL sp_customer_journey(10);` → customer 10's steps in time order
- [ ] `CALL sp_top_products(5);` → 5 best sellers with revenue and stock left
- [ ] `SELECT * FROM vw_cart_abandonment LIMIT 10;` → carted-but-unpaid items
- [ ] `SELECT * FROM vw_device_funnel;` → Mobile, Desktop, Tablet with cart → order %
- [ ] `UPDATE products SET stock_quantity = -1 WHERE product_id = 1636;` → refused by `chk_products_stock`
- [ ] `INSERT INTO reviews (customer_id, product_id, rating) VALUES (10, 1636, 9);` → refused by `chk_review_rating`

Visitor (not logged in)

- [ ] Home page → rotating 3D product ring, category tiles, trending products
- [ ] Shop all → 24 products per page, page numbers at the bottom
- [ ] Search "nike", filter Women, sort by price → matching results; Clear resets
- [ ] Open a product → photo, price, stock badge, reviews, "You may also like"
- [ ] Click Add to cart → sent to login; after logging in the item is in your cart
- [ ] Moon icon → dark mode; reload → still dark
- [ ] Phone width (or narrow window) → menu button, two products per row

Customer (`customer@example.com` / `pass123`)

- [ ] Register a new account; a wrong confirm password or a used email shows an error
- [ ] Register with the email `admin` → refused (security fix)
- [ ] Log in with a wrong password → "Incorrect password"
- [ ] Add 2 products; + and − change quantity; Remove deletes the line
- [ ] Press + past the stock → "Only N in stock"
- [ ] Checkout with an empty address → asked for one; with an address → payment page
- [ ] Pay with UPI → My orders shows the order as Paid; the product's stock fell
- [ ] Place another order, open it, click Cancel order → Cancelled; stock goes back
- [ ] Write a 4-star review → it appears with the average rating
- [ ] Profile shows your details; Log out

Seller (`seller@example.com` / `pass123` , or `trendstyle@gmail.com` / `seller123` after `set_seller_password`)

- [ ] Seller hub shows products listed, units sold, revenue
- [ ] Add a product with the ID left blank → next free ID used; it appears in the shop
- [ ] Add with an existing ID or text in Price → error message, nothing saved

- [ ] Edit price and stock → saved
- [ ] Delete a product that has orders → taken off sale (stock 0) instead

Admin (`admin` / `adminpass`)

- [ ] Overview: revenue, orders, customers, products, recent orders, low stock
- [ ] Orders: filter by status; Mark shipped, then Mark delivered; Cancel a paid order → stock returned, payment Refunded
- [ ] Products: search, Low stock filter, Edit, Funnel link
- [ ] Customers: search by name or city; Sellers: revenue per seller
- [ ] Funnel Analytics: change the dates → KPIs, funnel chart and daily chart update
- [ ] Product funnel: counts never rise from one stage to the next

Demo data

- [ ] `python manage.py generate_funnel_data` → about 300 shoppers; analytics shows a realistic funnel
- [ ] `python manage.py generate_funnel_data --delete` → demo data gone, stock restored

E. Setup and run

The full instructions are in the repository's README (github.com/Bhavesh-max-jpj/E-commerce-project). An existing database only needs the scripts it has not run yet; 03 to 09 are safe to re-run. A brand-new computer runs all of them in order.

1. Install: `pip install -r requirements.txt` (Django and PyMySQL).
2. Create `config/local_settings.py` with `DB_USER = "root"` and `DB_PASSWORD = "<your MySQL password>"`.
3. In MySQL Workbench, open and run (Execute All) each script, in this order:
   a. `database/01_schema.sql` : the 12 core tables with their keys and delete rules. **New computer only: it deletes and recreates them.**
   b. `database/02_seed_catalog.sql` : 16 sellers, 2,905 products. **New computer only.**
   c. `03_funnel_events.sql` → `04_triggers.sql` → `05_views_procedures.sql` → `06_constraints_cleanup.sql` → `07_device_view.sql` → `08_cumulative_funnel.sql` → `09_delete_rules.sql` . These are safe to run again.
4. In PowerShell, inside the project folder:
   a. `python manage.py migrate` (new computer only)
   b. `python manage.py create_sample_data`
   c. `python manage.py hash_plain_passwords`

   d. `python manage.py set_seller_password --all seller123` (optional)
   e. `python manage.py generate_funnel_data` (optional demo shoppers)
   f. `python manage.py runserver`
5. Open http://127.0.0.1:8000.

Who	Login page	Email or username	Password
Customer	/login/	customer@example.com	pass123
Seller	/seller/login/	seller@example.com	pass123
Seller with real products	/seller/login/	trendstyle@gmail.com	seller123 (after step 4.4)
Admin	/adminpanel/login/	admin	adminpass
Demo shoppers	/login/	any …@demo.shop	demo1234